

CLASS 9 · CBSE · GANITA MANJARI PART 2 · SHORT NOTES

# Chapter 13: Two Variables, One Line

Topic-wise short notes with solved examples

A linear equation in two variables has infinitely many solutions, and they form a straight line. This chapter covers the slope of a line and the ways of solving a pair of linear equations: substitution, elimination and graphs.

## 1 Linear equations in two variables

- An equation that can be written as  $ax + by + c = 0$ , where  $a, b, c$  are real numbers and  $a, b$  are not both zero, is a **linear equation in two variables**. This is its **standard form**.
- $a$  is the coefficient of  $x$ ,  $b$  is the coefficient of  $y$  and  $c$  is the constant.
- Fractions can be cleared by multiplying the whole equation by a suitable number.
- An equation in one variable is a special case:  $4x + 7 = 0$  is  $4x + 0 \cdot y + 7 = 0$ .

**Example 1.1** Write  $3x = 2y - 5$  and  $4y = 9$  in standard form and state  $a, b$  and  $c$ .

**SOLUTION**

$$3x - 2y + 5 = 0 \text{ and } 0 \cdot x + 4y - 9 = 0$$

$$a = 3, b = -2, c = 5; \text{ and } a = 0, b = 4, c = -9$$

**Example 1.2** A fruit seller sells mangoes at ₹80 per kg and bananas at ₹50 per kg. A customer buys  $x$  kg of mangoes and  $y$  kg of bananas and pays ₹340. Write an equation.

**SOLUTION**

$$\text{Cost of mangoes} = 80x \text{ and cost of bananas} = 50y$$

$$80x + 50y = 340$$

## 2 Solutions and their graph

- A **solution** is a pair of values  $(x, y)$  that satisfies the equation. It is an **ordered pair**: the  $x$ -value is written first.
- A linear equation in two variables has **infinitely many** solutions. Choose any value of  $x$  and solve for  $y$ .
- Plotted as points, all the solutions lie on one **straight line**, and every point of that line is a solution.
- Easy points: put  $x = 0$  to find where the line cuts the  $y$ -axis, and  $y = 0$  to find where it cuts the  $x$ -axis.
- If  $c = 0$ , the line  $ax + by = 0$  passes through the origin.

- Multiplying an equation by a non-zero number does not change its solutions:  $3x + 4y = 7$  and  $6x + 8y = 14$  are the same line.

**Example 2.1** Check whether  $(3, 2)$  and  $(2, 3)$  are solutions of  $2x + 3y = 12$ . Find the points where its graph cuts the axes.

**SOLUTION**

$(3, 2)$ :  $6 + 6 = 12$ .  $(2, 3)$ :  $4 + 9 = 13 \neq 12$ .

$x = 0$  gives  $y = 4$ , and  $y = 0$  gives  $x = 6$ .

**$(3, 2)$  is a solution and  $(2, 3)$  is not. The line cuts the axes at  $(0, 4)$  and  $(6, 0)$ .**

**Example 2.2**  $(2, -1)$  is a solution of both  $3ax + 2y = 10$  and  $x - by = 5$ . Find  $a$  and  $b$ .

**SOLUTION**

First equation:  $6a - 2 = 10$ , so  $a = 2$ .

Second equation:  $2 + b = 5$ , so  $b = 3$ .

**$a = 2, b = 3$**

### 3 Slope of a line

- The **slope** (gradient) measures the steepness of a line. Moving from one point of the line to another, the horizontal change is the **run** and the vertical change is the **rise**.

$$\text{Slope} = \frac{\text{Rise}}{\text{Run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

- The slope is the same whichever two points of the line are used, and in whichever order.

| Slope     | The line                                    |
|-----------|---------------------------------------------|
| Positive  | rises from left to right                    |
| Negative  | falls from left to right                    |
| Zero      | is horizontal, such as $y = 5$              |
| Undefined | is vertical, such as $x = 2$ (the run is 0) |

**Example 3.1** Find the slope of the line through (a)  $(1, 2)$  and  $(5, 10)$  (b)  $(-2, 7)$  and  $(4, -5)$ .

**SOLUTION**

$$(a) \frac{10 - 2}{5 - 1} = \frac{8}{4} \quad (b) \frac{-5 - 7}{4 - (-2)} = \frac{-12}{6}$$

**(a) 2 (b) -2**

**Example 3.2** A ramp rises 1.5 m over a horizontal length of 6 m. Find its slope. How long must a ramp of the same slope be to rise 2 m?

SOLUTION

$$\text{Slope} = \frac{1.5}{6} = 0.25$$

$$\text{Run} = \frac{2}{0.25}$$

**Slope 0.25; a horizontal length of 8 m**

## 4 Slope-intercept form

- $y = mx + d$  is the **slope-intercept form**.  $m$  is the slope and  $d$  is the **y-intercept**: the line cuts the  $y$ -axis at  $(0, d)$ .
- Why  $m$  is the slope: for two points on the line,  $y_2 - y_1 = (mx_2 + d) - (mx_1 + d) = m(x_2 - x_1)$ .
- From the standard form  $ax + by + c = 0$  with  $b \neq 0$ :  $y = -\frac{a}{b}x - \frac{c}{b}$ , so

$$m = -\frac{a}{b} \quad d = -\frac{c}{b}$$

- Two lines with the same slope and different  $y$ -intercepts are parallel.

**Example 4.1** Find the slope and the  $y$ -intercept of  $4x + 2y = 10$ .

SOLUTION

$$2y = -4x + 10, \text{ so } y = -2x + 5.$$

**Slope  $-2$ ;  $y$ -intercept 5. For each increase of 1 in  $x$ ,  $y$  falls by 2.**

## 5 Pair of linear equations

- Two linear equations in the same two variables form a **pair**:  $a_1x + b_1y + c_1 = 0$  and  $a_2x + b_2y + c_2 = 0$ .
- A **solution of the pair** is an ordered pair that satisfies **both** equations.
- The idea of every method is to reduce the pair to one equation in one variable.

## 6 Substitution method

- **Step 1**: From one equation, express one variable in terms of the other.
- **Step 2**: Substitute this expression into the other equation and solve.
- **Step 3**: Put the value back to find the other variable. Check in both equations.

**Example 6.1** Solve  $2x + 3y = 13$  and  $x - y = 4$ .

**SOLUTION**

From the second equation,  $x = y + 4$ .

Substitute in the first:  $2(y + 4) + 3y = 13$ , so  $5y = 5$  and  $y = 1$ .

Then  $x = 1 + 4 = 5$ . Check:  $10 + 3 = 13$  and  $5 - 1 = 4$ .

$x = 5, y = 1$

**Example 6.2** Two angles of a triangle are  $x^\circ$  and  $y^\circ$  with  $y = 3x$ , and the third angle is  $60^\circ$ . Find  $x$  and  $y$ .

**SOLUTION**

$x + y + 60 = 180$ . Substituting  $y = 3x$ :  $4x = 120$ .

$x = 30, y = 90$

## 7 Elimination method

- **Step 1:** Multiply the equations by suitable numbers so that one variable has the same coefficient in both.
- **Step 2:** Add or subtract the equations to remove that variable, and solve.
- **Step 3:** Put the value back to find the other variable.

**Example 7.1** Solve  $3x + 2y = 16$  and  $5x - 3y = 14$ .

**SOLUTION**

Multiply the first by 3 and the second by 2:  $9x + 6y = 48$  and  $10x - 6y = 28$ .

Adding:  $19x = 76$ , so  $x = 4$ . Then  $12 + 2y = 16$  gives  $y = 2$ .

$x = 4, y = 2$

**Example 7.2** 3 pens and 2 notebooks cost ₹95. 2 pens and 5 notebooks cost ₹155. Find the cost of each.

**SOLUTION**

Let a pen cost ₹ $x$  and a notebook ₹ $y$ :  $3x + 2y = 95$  and  $2x + 5y = 155$ .

Multiply by 2 and by 3:  $6x + 4y = 190$  and  $6x + 15y = 465$ . Subtracting:  $11y = 275$ , so  $y = 25$ .

Then  $3x = 95 - 50$ , so  $x = 15$ .

**A pen costs ₹15 and a notebook costs ₹25.**

**Example 7.3** The digits of a two-digit number add up to 11. The number formed by reversing the digits is 27 more than the number. Find the number.

**SOLUTION**

Let the tens digit be  $x$  and the units digit be  $y$ . The number is  $10x + y$  and the reversed number is  $10y + x$ .

$x + y = 11$  and  $(10y + x) - (10x + y) = 27$ , that is  $y - x = 3$ .

Adding:  $2y = 14$ , so  $y = 7$  and  $x = 4$ .

**The number is 47.**

**Example 7.4** Solve  $7x + 5y = 46$  and  $5x + 7y = 50$  by adding and subtracting.

**SOLUTION**

The coefficients are interchanged. Adding:  $12x + 12y = 96$ , so  $x + y = 8$ .

Subtracting:  $2x - 2y = -4$ , so  $x - y = -2$ .

$x = 3, y = 5$

## 8 How many solutions?

- During elimination, both variables may disappear together. What is left decides the answer.

| Condition                                                | Number of solutions | The two lines        |
|----------------------------------------------------------|---------------------|----------------------|
| $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$                   | exactly one         | intersect at a point |
| $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$    | infinitely many     | coincide             |
| $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ | none                | are parallel         |

- In the second case one equation is a multiple of the other, and elimination gives  $0 = 0$ . In the third case elimination gives a false statement such as  $0 = 1$ .

**Example 8.1** Without solving, find the number of solutions of (a)  $2x + 3y = 7, 4x + 6y = 14$  (b)  $2x + 3y = 7, 4x + 6y = 10$  (c)  $x + 2y = 5, 3x - y = 1$ .

**SOLUTION**

(a)  $\frac{2}{4} = \frac{3}{6} = \frac{7}{14}$  (b)  $\frac{2}{4} = \frac{3}{6} \neq \frac{7}{10}$  (c)  $\frac{1}{3} \neq \frac{2}{-1}$

**(a) infinitely many (b) none (c) exactly one**

**Example 8.2** For what value of  $k$  does the pair  $kx + 6y = 9, 2x + 3y = 5$  have no solution?

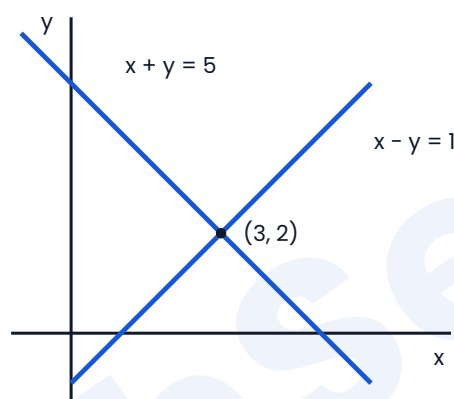
**SOLUTION**

We need  $\frac{k}{2} = \frac{6}{3} \neq \frac{9}{5}$ . The first two give  $k = 4$ , and  $2 \neq \frac{9}{5}$ .

$$k = 4$$

## 9 Graphical method

- Draw the two lines on the same axes. The solutions of the pair are the points that lie on **both** lines.



The lines  $x + y = 5$  and  $x - y = 1$  meet at  $(3, 2)$

- Lines that **intersect** give one solution, the point of intersection. **Parallel** lines give no solution. **Coincident** lines give infinitely many.

**Example 9.1** Solve  $x + y = 5$  and  $x - y = 1$  graphically.

**SOLUTION**

$x + y = 5$  passes through  $(0, 5)$  and  $(5, 0)$ .  $x - y = 1$  passes through  $(1, 0)$  and  $(0, -1)$ .

The two lines cross at  $(3, 2)$ . Check:  $3 + 2 = 5$  and  $3 - 2 = 1$ .

$$x = 3, y = 2$$

**Example 9.2** Plan A costs ₹100 a month plus ₹2 per GB. Plan B costs ₹40 a month plus ₹5 per GB. For what usage do the two plans cost the same?

**SOLUTION**

For  $x$  GB the costs are  $y = 100 + 2x$  and  $y = 40 + 5x$ .

$100 + 2x = 40 + 5x$  gives  $x = 20$ , and then  $y = 140$ .

**At 20 GB both cost ₹140. Below 20 GB Plan B is cheaper, and above it Plan A is cheaper.**

## 10 Common mistakes

- Writing a solution in the wrong order:  $(3, 2)$  means  $x = 3$  and  $y = 2$ .
- Subtracting in different orders in the slope formula. Use  $y_2 - y_1$  over  $x_2 - x_1$  with the same point first in both.
- Reading the slope of  $4x + 2y = 10$  as 4. First write it as  $y = -2x + 5$ .
- Multiplying only one side of an equation when preparing for elimination.
- Sign errors when subtracting one equation from another. Change the sign of every term.
- Comparing the ratios before writing both equations in the same form.

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